The Ceiling of Cooperation in Extrinsic Update Rules

Harry Foster, Vince Knight, Sebastian Krapohl

June 24, 2026

Abstract

Evolutionary game theory provides a framework by which to study the emergence of cooperation in a population of self-interested actors. In such a framework, players' decisions on whether or not to cooperate evolve according to decision rules called population dynamics. However, often games are studied under the assumption that all individuals play under the same conditions, and many common choices of update rule are not well suited for a heterogeneous population. In this paper, we categorise and compare four different population dynamics in such a population as "extrinsic", where players learn by looking outward at the payoffs of other players, and "intrinsic", where players look inwardly at their own attributes or potential payoffs. We show that extrinsic population dynamics admit a ceiling on the rate of cooperation which can be exceeded by intrinsic population dynamics, and demonstrate this using the public goods game with heterogeneous contributions.

1 Introduction

The conditions under which cooperation can emerge among self-interested actors is a central question across biology, economics, and the social sciences. Many of the situations in which this question arises are collective action problems, in which individuals share an interest in a common good yet each would prefer that others bear the cost of providing it. The structure of such a dilemma maps onto a *public goods game* [3, 21, 22, 23, 47]: each actor decides whether to incur a private cost to contribute to a shared resource whose total value is multiplied by some factor r and distributed equally amongst all players. This creates the temptation to free-ride, and so purely selfish reasoning leads to a tragedy of the commons [20, 56], even when collective cooperation would benefit everyone. Both theory [45] and controlled experiments [34, 35] show that framing cooperation as a collective-risk problem, where failure to meet a cooperation threshold incurs a shared loss, can substantially raise cooperation rates.

Evolutionary game theory offers a framework for studying how cooperation can emerge among self-interested actors without assuming rational foresight [24, 37]. In this approach, strategies that perform well spread through a population via one of several update rules, while poorly performing strategies decline. The conditions under which cooperative strategies can invade and stabilise have been extensively studied in two-player settings such as the iterated prisoner's dilemma [5, 28] and the snowdrift game [55]. The public goods game generalises these pairwise dilemmas to N players [21], making it well suited to studying multi-lateral cooperation. In finite populations, evolutionary outcomes are stochastic: a rare mutant strategy spreads or vanishes with a probability that depends on the population size, payoff structure, and update rule [39, 53].

The conditions under which one strategy invades a population have been studied in depth for particular evolutionary processes. The $\frac{1}{N}$ fixation probability of a single mutant under neutral drift [37] serves as a benchmark for when selection favours a strategy, and results such as the $\frac{1}{3}$ rule [6, 40] hold even when a population updates with both the Moran process and Fermi imitation dynamics [31]. For symmetric 2×2 games, the condition $a(N-2) + bN > cN + d(N-2)$ [1, 26] on the payoff entries makes one strategy more abundant than the other, and this holds across a wide range of evolutionary processes and for arbitrary mutation and selection intensity [2, 17]. These results, however, concern a symmetric game played pairwise between identical members of a population. We instead consider all N-player games in which a defector outperforms a cooperator in a given state, amongst a fully heterogeneous population.

A feature of real-world collective action that standard models typically set aside is that actors are not identical. Players differ in wealth, productive capacity, and the costs they face, and this inequality can reshape the dynamics of cooperation [23]. In spatial and graphical settings, where players sit on a network and participate simultaneously in multiple games with their neighbours [38, 41, 50], heterogeneity generally promotes cooperation. In [46] it is shown that social diversity in endowments promotes the emergence of cooperation in the public goods game directly: heterogeneous populations outperform their homogeneous counterparts over a wide range of parameters. This echoes earlier work showing that scale-free network topology, a form of structural heterogeneity, provides a unifying framework for the emergence of cooperation [44]. When contributions depend on dynamically accumulated reputation [33], high-reputation defectors receive reduced transfers, weakening defection's advantage. When heterogeneity arises from differences in network degree so that players allocate contributions according to the strength of their social ties [29], the payoff structure penalises high-degree defectors and protects cooperating clusters from invasion. The characteristic spatial mechanism is the formation of cooperating clusters that are more profitable than their defecting neighbours; provided r is sufficiently large, these clusters resist invasion and eventually spread. Heterogeneity does not, however, unconditionally favour cooperation. Flores et al. [11] show that when the contribution level is itself a strategic variable rather than a fixed attribute, low-value contributions act as a form of defection against the most collectively beneficial strategies, so that the emergence of cooperation does not guarantee the emergence of high-value cooperation.

How the choice of update rule interacts with population heterogeneity is also poorly understood. The Moran process [27, 36], Fermi imitation dynamics [51], aspiration dynamics [10], and introspection dynamics [7,8] all model strategy revision in distinct ways, and each may respond differently to player-specific payoffs. In particular, *intrinsic* dynamics, in which a player evaluates a prospective strategy change based on their payoff in comparison with their own counterfactual payoff or aspiration level, are better suited to heterogeneous populations than *extrinsic* dynamics, in which a player may copy a high-fitness neighbour without considering whether the same strategy would serve them as well.

We make the following contributions. First, we build on an established general framework for evolutionary dynamics on fully heterogeneous, well-mixed populations of N ordered individuals, in which each player may have a distinct payoff function and contribution level. Second, we formalise the notion of a purely extrinsic population dynamic, a definition encompassing both the Moran process [27, 36] and Fermi imitation dynamics [51], and show that such dynamics are inherently constrained by an upper bound on the probability of cooperation. Third, we consider two purely intrinsic population dynamics, aspiration dynamics [10] and introspection dynamics [7,8], showing that these processes can overcome the ceiling of purely extrinsic dynamics. For each population dynamic, we use the public goods game [3, 21, 22, 23, 47] to demonstrate our results.

The remainder of the paper is organised as follows. Section 2 introduces the population model and the public goods game. Section 3 introduces a formal definition of a *purely extrinsic population dynamic*, gives formulations of the Moran process and Fermi imitation dynamics in terms of our population model, proves that these satisfy the definition of a purely extrinsic population dynamic, and proves the ceiling on the probability of cooperation in such dynamics. We then demonstrate this using a heterogeneous public goods game. Section 4 gives formulations of two intrinsic population dynamics and shows that they can exceed the ceiling on the probability of cooperation, demonstrated using their results in the public goods game. Finally, Section 5 confirms through direct simulation of the Markov chain that these results persist for large populations.

2 The general heterogeneous population dynamic model

As in [7,8], we define a general framework for evolutionary dynamics on a fully heterogeneous, well-mixed population.

2.1 The state space

We model a population consisting of N **ordered** individuals. Each individual selects an action from a common action set $A = (A_1, A_2, \ldots, A_k)$ of size k. A *state* of the population is an ordered N-tuple $\mathbf{a} = (a_1, a_2, \ldots, a_N) \in A^N$, and the *state space* S is the set of all such tuples. A payoff function $\pi\colon S \to \mathbb{R}^N$ assigns to each state a vector of individual payoffs; we write $\pi_i(\mathbf{a})$ for the payoff of individual i in state $\mathbf{a}$.

The state space of an evolutionary game can be described as the graph whose nodes are the states $\mathbf{a} \in S$, with an edge between two vertices $\mathbf{a}, \mathbf{b}$ if and only if they differ in exactly one position. This defines an evolutionary process in which one player changes action type at each time step. Each state also has a self-loop, as at a given time step, a player may accept the same strategy they currently possess, or reject the strategy which they have considered. We call the unique coordinate at which $\mathbf{a}$ and $\mathbf{b}$ differ the *index of difference*, denoted $I(\mathbf{a}, \mathbf{b})$. The set of states reachable from $\mathbf{a}$ in one step is denoted $\text{Neb}(\mathbf{a}) = \{\mathbf{b} \in S : h(\mathbf{a}, \mathbf{b}) = 1\}$. This notation follows [8].

In the case of a game with two actions, the state space of the Markov chain is analogous to an N-dimensional hypercube. This is the graph whose node set consists of the 2^N N-dimensional boolean vectors, and which has edges between vertices if and only if they differ in exactly one position [19]. The state space of an evolutionary game with two actions can be modelled in this way by denoting the strategies 0 and 1. Throughout, the two-action games we study are social dilemmas, so we take these two actions to be cooperate (C) and defect (D) and write the state space as $\{C, D\}^N$. The graphical representation of a birth-death process differs from a hypercube in exactly two ways. First, a population may remain the same at a given step, and thus each vertex contains a self-loop. Second, the probability of transitioning from a state $\mathbf{a}$ to a state $\mathbf{b}$ is not necessarily the same as the probability of transitioning from $\mathbf{b}$ to $\mathbf{a}$. Thus, instead of one edge between two states, we have two directed edges, $\mathbf{a} \to \mathbf{b}$ and $\mathbf{b} \to \mathbf{a}$. Figure 2(a) and 2(b) illustrate the population model and its state space.

2.2 Population dynamics and mutation

The population evolves as a Markov chain [48] on S. We therefore obtain a transition matrix T, where $T_{\mathbf{ab}}$ denotes the probability of transitioning from a state $\mathbf{a}$ to a state $\mathbf{b}$, as shown in figure 2(c). They take the general form:

$$T_{\mathbf{ab}} = \begin{cases} P(\mathbf{a} \to \mathbf{b}) & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}), \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b}. \end{cases} \tag{1}$$

where $P(\mathbf{a} \to \mathbf{b})$ is the probability that player $I(\mathbf{a}, \mathbf{b})$ changes from its action type in $\mathbf{a}$ to its action type in $\mathbf{b}$. The process which determines this probability is called a *population dynamic.* We shall discuss four population dynamics in this paper, and classify them as shown in Figure 1. Purely extrinsic population dynamics only consider the success of a strategy in the current population when updating a player's action type. This models a behaviour where players look outwardly to attempt to improve their payoff. Purely intrinsic population dynamics model a player who looks inwardly to improve their payoff, comparing their current payoff to some value which does not depend on the payoffs of an action type in the current population. The dynamics discussed in this paper are shown in figure 1

Two of the dynamics discussed in this paper, the Moran process and Fermi imitation dynamics, correspond to absorbing Markov chains, where the measurable outcome is the probability of absorption into a given absorbing state. The remaining two, introspection and aspiration dynamics, are ergodic chains, for which we can compute the long-run abundance of a given strategy in the population. To enable meaningful comparison across all models, we introduce mutation. Mutation ensures that all processes become ergodic, while also introducing a controlled amount of noise: at any moment, individuals may switch to a different strategy [17]. We assume a mutation parameter $\mu \in [0,1]_{\mathbb{R}}^{N \times k}$, where μ_{ij} denotes the probability that individual i adopts strategy j via mutation. The stationary abundance of a strategy C, denoted p_C, is well defined only with mutation for the Moran process and Fermi imitation dynamics, so we report these for $\mu > 0$; the introspection and aspiration dynamics are ergodic for every $\mu \geq 0$. The figures throughout use the mutation rates $\mu \in \{0.001, 0.005, 0.05, 0.1\}$.

Given a process defined by transition matrix T and two states $\mathbf{a}$ and $\mathbf{b}$, the probability of the original process *with mutation* transitioning from $\mathbf{a}$ to $\mathbf{b}$ is given by:

$$P^{(\mu)}(\mathbf{a} \to \mathbf{b}) = P(\text{pick individual } I(\mathbf{a},\mathbf{b})) \cdot \big(P(\mathbf{a}_{I(\mathbf{a},\mathbf{b})} \text{ transitions} \to \mathbf{b}_{I(\mathbf{a},\mathbf{b})}) \cdot P(\text{no mutation}) + P(\mathbf{a}_{I(\mathbf{a},\mathbf{b})} \text{ mutates} \to \mathbf{b}_{I(\mathbf{a},\mathbf{b})})\big)$$

For a focal individual i, the probability that no mutation occurs is $1 - \sum_{j=1}^{k} \mu_{ij}$. Let T be the transition matrix for a population dynamic and $T_{\mathbf{ab}}$ be the transition probability from a state $\mathbf{a}$ to a state $\mathbf{b}$. Then the transition matrix $T^{(\mu)}$ for the process with mutation is given by:

$$T^{(\mu)}_{\mathbf{ab}} = \begin{cases} T_{\mathbf{ab}} \left(1 - \sum_{j=1}^{k} \mu_{I(\mathbf{a},\mathbf{b}),j}\right) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),\mathbf{b}_{I(\mathbf{a},\mathbf{b})}}}{N} & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}), \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b}. \end{cases} \tag{2}$$

This behaviour is shown in figure 2(d). Note that if $\sum_{j=1}^{k} \mu_{ij} = 1$, mutation fully determines the behaviour of individual i. This construction ensures that all processes are ergodic, rather than absorbing. Consequently, we no longer obtain absorption matrices for any model with mutation; instead, we derive a unique steady state that is independent of the initial state. All processes in this paper are considered under the effect of action-invariant mutation, where $\mu_{iw} = \mu_{il}$ for all action types w, l. This ensures that mutation does not cause an inherent bias towards a given action type.

3 Extrinsic population dynamics

We begin by defining a *purely extrinsic population dynamic.* Intuitively, a purely extrinsic process revises strategies through payoff comparison between members of a population.

Definition 1 (Purely extrinsic population dynamic)**.** *Let T be the transition matrix of a population dynamic on a state space S. We say that T defines a* purely extrinsic *process if for every $\mathbf{a}, \mathbf{b} \in S$ with $\mathbf{b} \in$ Neb$(\mathbf{a})$, $T_{\mathbf{ab}}$ satisfies:*

1. *$T_{\mathbf{ab}}$ is a function of payoffs evaluated in $\mathbf{a}$ and in no other state.*

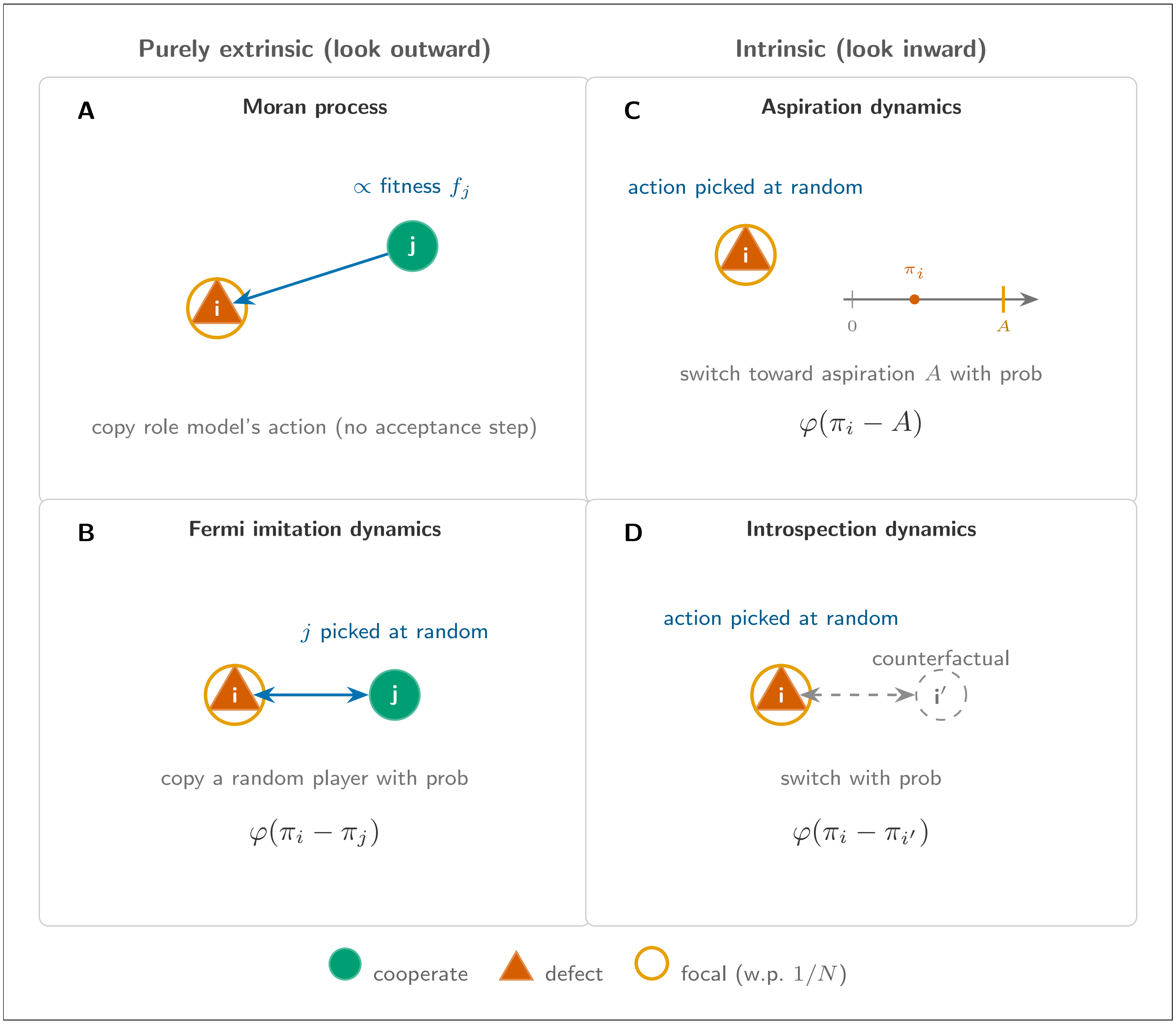

Figure 1: **The four population dynamics.** In each panel the triangle is chosen for replacement with probability $\frac{1}{N}$ and the diamond is chosen with a probability that depends on the process; the filled shape is the action that may be adopted. **A,** the Moran process and **B,** Fermi imitation dynamics are purely extrinsic. **C,** aspiration dynamics and **D,** introspection dynamics are purely intrinsic. For introspection dynamics the candidate action is chosen with probability $\frac{1}{k-1}$, and for aspiration dynamics there is only one other action to accept.

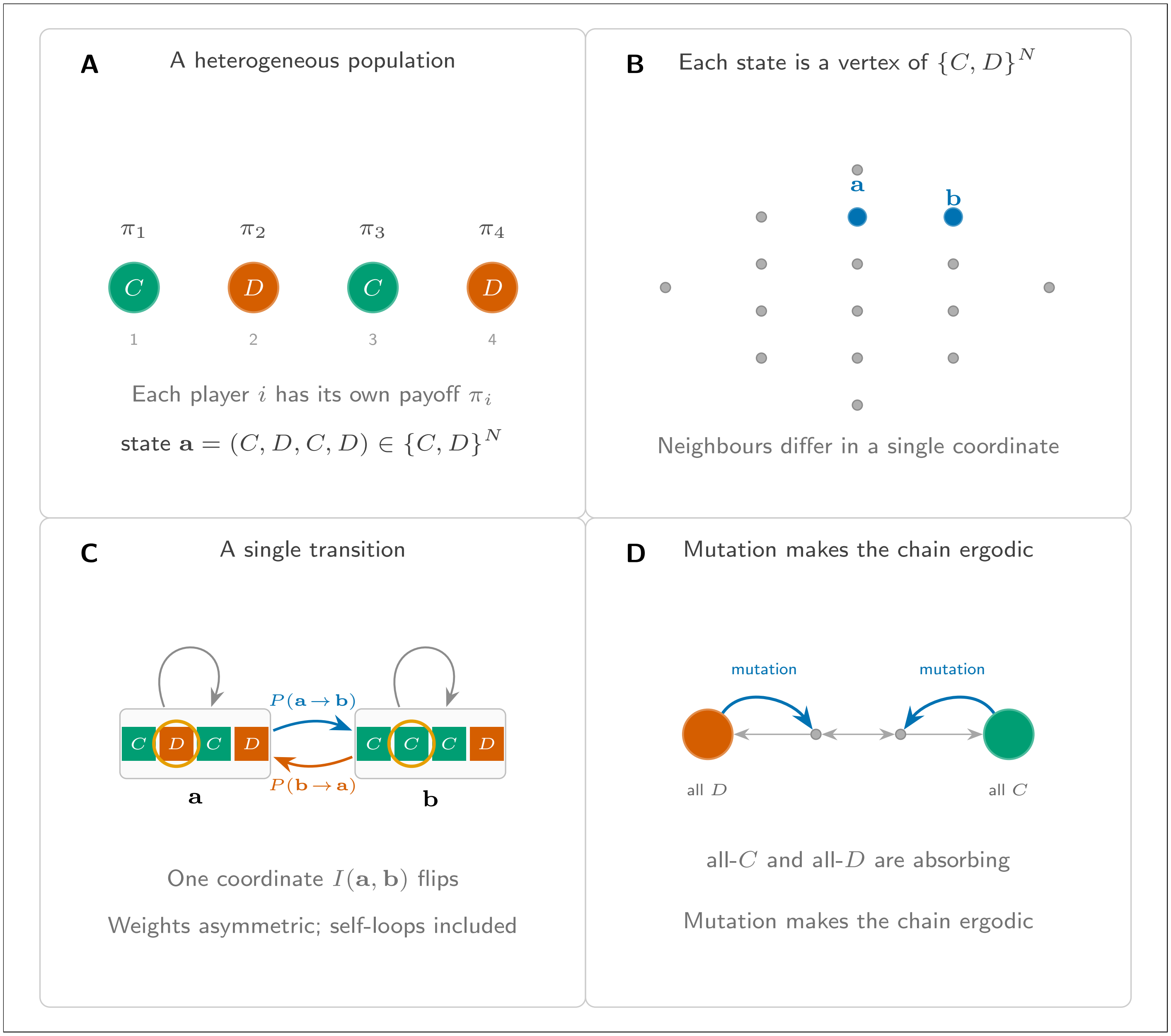

Figure 2: **The heterogeneous population as a Markov chain. A,** A heterogeneous population with action set $A = \{C, D\}$, in which each player has its own payoff function. **B,** Each state is a vertex of $S = A^N$, so the state space is an N-dimensional hypercube. **C,** A single transition between neighbouring states. **D,** Mutation connects the chain and makes it ergodic.

2. *Let $\mathcal{R}$ denote the set of players j such that $j \neq I(\mathbf{a}, \mathbf{b})$ and $\mathbf{a}_j = \mathbf{b}_{I(\mathbf{a},\mathbf{b})}$. Then $T_{\mathbf{ab}}$ is non-decreasing in the payoffs $\pi_j(\mathbf{a})$ for all $j \in \mathcal{R}$ (and strictly increasing in at least one such j if $\mathcal{R} \neq \emptyset$), and non-increasing in the payoffs $\pi_j(\mathbf{a})$ for all $j \notin \mathcal{R}$.*

3. *$T_{\mathbf{ab}}$ contains no term dependent on $a_{I(\mathbf{a},\mathbf{b})}$ or $b_{I(\mathbf{a},\mathbf{b})}$, except in the payoff function $\pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a})$, mutation parameters μ_{ij}, and in the definition of the set $\mathcal{R}$.*

Condition (1) gives the property of purely extrinsic population dynamics that looks outwardly, copying other individuals in the state. Condition (2) captures the basic monotonicity: a higher payoff for a role model makes that role model more likely to be copied, and a higher p;ayoff for a non-role-model pulls the other way. Condition (3) rules out an inherent bias towards either action. Without (3), we must consider population dynamics which explicitly favour a certain action type, for example:

$$\bar{T}_{\mathbf{ab}} = \begin{cases} \delta_{I(\mathbf{a},\mathbf{b})} \cdot T_{\mathbf{ab}} & \text{if } \mathbf{b} \neq \mathbf{a}, \\ 1 - \sum_{\mathbf{c} \neq \mathbf{a}} T_{\mathbf{ac}} & \text{if } \mathbf{b} = \mathbf{a}. \end{cases}$$

where $T_{\mathbf{ab}}$ is the transition matrix of some purely extrinsic dynamic, and $\delta_{I(\mathbf{a},\mathbf{b})}$ takes the value 1 if $b_{I(\mathbf{a},\mathbf{b})} = C$ and $\frac{1}{2}$ if not. Without (3), $\bar{T}$ would define a purely extrinsic population dynamic, but would be tailored to favour a particular action type. We exclude cases such as this, as they do not conform to the framework in which fitness is the single measure of the success of action types in the population, and thus of which strategy is passed on in each step of the process. While parameters such as choice intensity and selection intensity affect the transition probability $T_{\mathbf{ab}}$, they do not inherently favour any specific action type.

We now show two classical examples of purely extrinsic population dynamics: the Moran process, in which a player is chosen to be duplicated proportional to their payoff, and Fermi imitation dynamics, which directly compares the payoffs of pairs of players.

3.1 The Moran process

The Moran process [36] is defined by the following algorithm:

1. A player is chosen to be copied with a probability proportional to their fitness in the population
2. A player is chosen with probability $\frac{1}{N}$ to change their strategy

This is shown in Figure 1A, and the transition matrix is given by:

$$T_{\mathbf{ab}} = \begin{cases} \dfrac{\sum_{j:\, a_j = b_{I(\mathbf{a},\mathbf{b})}} f_j(\mathbf{a})}{N \sum_j f_j(\mathbf{a})} \left(1 - \sum_{j=1}^{k} \mu_{I(\mathbf{a},\mathbf{b}),j}\right) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),\mathbf{b}_{I(\mathbf{a},\mathbf{b})}}}{N} & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}), \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b}. \end{cases} \tag{3}$$

Where $f_i(\mathbf{a}) = 1 - \varepsilon + \varepsilon \pi_i(\mathbf{a})$ is the fitness of a player i in state $\mathbf{a}$ scaled with selection intensity $\varepsilon > 0$. The value of ε is chosen to ensure a positive fitness for all players in order that the probability $T_{\mathbf{ab}}$ is always well defined. This parameter controls the rationality with which players choose which strategy to adopt in each time step: a larger ε indicates that a player will choose a higher fitness strategy with a greater probability.

The Moran process has often been employed as a model of biological evolution [37], where a player's "type" represents a species or allele within a population. In each time step, rather than a given player choosing a new action type, a randomly chosen individual in the population would die and be removed from the population. At the same time, an individual would reproduce with a probability proportional to their fitness, introducing an exact copy of themselves to the population to replace the removed individual. As a result of the model's original purpose, the Moran process contains no decision step where a player chooses whether or not to accept a considered strategy. Instead, at each time step, a player *must* accept another player's action, and we remain in the same state when a player accepts the same action as they currently play.

One implication of this is that in a heterogeneous population, worsening moves may be taken with a high probability, as an action that provides a high payoff for one player in a population may not provide the same return for another. In nature, an offspring is unable to choose its parents, and thus cannot choose which attributes it is born with, and so the Moran process does not model the decision which players have regarding whether or not to play a certain strategy. When modelling the adoption of behaviours, these assumptions may lead to players making decisions which are harmful for their own payoff.

We now show that the Moran process satisfies the conditions of Definition 1.

Theorem 1 (The Moran process is a purely extrinsic process)**.** *The Moran process, as defined by Equation* (3)*, is a purely extrinsic population dynamic.*

Proof. By definition the Moran process satisfies conditions (1) and (3). It remains to be shown that $T_{\mathbf{ab}}$ is non-decreasing in $\pi_j(\mathbf{a})$ for $j \in \mathcal{R}$ (strictly increasing for at least one such j) and non-increasing in $\pi_j(\mathbf{a})$ for $j \notin \mathcal{R}$. Recall $f_i(\mathbf{a}) = 1 - \varepsilon + \varepsilon\pi_i(\mathbf{a})$ for some selection intensity $\varepsilon > 0$. Then we have:

$$T_{\mathbf{ab}} = \frac{\sum_{j\in\mathcal{R}} f_j(\mathbf{a})}{N\sum_{l=1}^{N} f_l(\mathbf{a})}\left(1 - \sum_{i=1}^{k}\mu_{I(\mathbf{a},\mathbf{b}),i}\right) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),b_{I(\mathbf{a},\mathbf{b})}}}{N}$$

$$\frac{dT_{\mathbf{ab}}}{d\pi_i(\mathbf{a})} = \frac{\varepsilon\sum_{x\notin\mathcal{R}} f_x(\mathbf{a})}{\left(N\sum_{l=1}^{N} f_l(\mathbf{a})\right)^2}\left(1 - \sum_{i=1}^{k}\mu_{I(\mathbf{a},\mathbf{b}),i}\right) \quad \text{for } i \in \mathcal{R}$$

For $i \in \mathcal{R}$, as ε is chosen such that $f_i(\mathbf{a}) > 0$ for all $i \in \mathbf{a}$, and $\mu_{i,j}$ such that $\sum_{i=1}^{k}\mu_{I(\mathbf{a},\mathbf{b}),i} < 1$, this derivative is strictly positive. For $j \notin \mathcal{R}$, the analogous computation gives $\frac{dT_{\mathbf{ab}}}{d\pi_j(\mathbf{a})} = -\frac{\varepsilon\sum_{x\in\mathcal{R}} f_x(\mathbf{a})}{(N\sum_{l=1}^{N} f_l(\mathbf{a}))^2}(1 - \sum_{i=1}^{k}\mu_{I(\mathbf{a},\mathbf{b}),i}) \leq 0$, so condition (2) holds. ☐

3.2 Fermi imitation dynamics

Fermi imitation dynamics [51] is another commonly employed method of modelling evolution in a population. It follows the process:

- Two players, i and j are selected at random from the population.
- Player i copies the action type of the player j with a probability according to the Fermi function $\phi(\pi_i(\mathbf{a}) - \pi_j(\mathbf{a})) = \frac{1}{1+e^{(\beta(\pi_i(\mathbf{a})-\pi_j(\mathbf{a})))}}$

Where β is the *choice* intensity of the system, representing how sensitive our players' choice is to the difference between their current payoff and the payoff which they are considering. This algorithm is also shown in Figure 1B. This method has been commonly used in studies of evolutionary games, for example [29, 33, 51]. We now define our transition matrix according to the Fermi function:

$$T_{\mathbf{ab}} = \begin{cases} \dfrac{1}{N(N-1)}\sum_{j:\, a_j = b_{I(\mathbf{a},\mathbf{b})}} \phi\big(\pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a}) - \pi_j(\mathbf{a})\big)\left(1 - \sum_{j=1}^{k}\mu_{I(\mathbf{a},\mathbf{b}),j}\right) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),b_{I(\mathbf{a},\mathbf{b})}}}{N} & \text{if } \mathbf{b} \in \mathrm{Neb}(\mathbf{a}), \\ 0 & \text{if } \mathbf{b} \notin \mathrm{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b}\in S\setminus\{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b}. \end{cases} \tag{4}$$

Unlike a Moran Process, we can see that the players are not selected with a probability proportional to their fitness. Instead, Fermi Imitation Dynamics selects a pair of players and compares their payoffs directly, giving players the option not to accept a strategy at a given time step. However, similarly to the Moran process, in a heterogeneous population the method of adopting a new strategy based entirely on the payoff of another individual may lead to worsening moves in which players adopt strategies which currently admit a higher fitness, but which will not improve their own fitness.

We also see that due to the formulation of the logit function ϕ, Fermi imitation dynamics can become invariant under certain parameters. This occurs in any game in which one action type's payoff in any given state can be expressed as another action type's payoff up to the addition of some term γ, which depends only on a subset of the game's parameters. In that case, $\Delta(\pi) = \pm\gamma$, and is therefore invariant to any parameters of the game of which γ is not a function.

We now show that Fermi imitation dynamics satisfies the definition of a purely extrinsic population dynamic.

Theorem 2 (Fermi imitation dynamics is a purely extrinsic population dynamic)**.** *Fermi imitation dynamics, as defined by Equation* (4)*, is a purely extrinsic population dynamic.*

Proof. By definition, Fermi imitation dynamics satisfies conditions (1) and (3). It remains to show that it satisfies condition (2).

$$T_{\mathbf{ab}} = \frac{1}{N(N-1)}\sum_{j\in\mathcal{R}}\phi(\pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a}) - \pi_j(\mathbf{a}))\left(1 - \sum_{i=1}^{k}\mu_{I(\mathbf{a},\mathbf{b}),i}\right) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),b_{I(\mathbf{a},\mathbf{b})}}}{N}$$

As ϕ is strictly decreasing, every term in the sum is strictly increasing in $\pi_j(\mathbf{a})$ for $j \in \mathcal{R}$. For $j \notin \mathcal{R}$ with $j \neq I(\mathbf{a}, \mathbf{b})$, $\pi_j(\mathbf{a})$ does not appear in the formula at all. For $j = I(\mathbf{a}, \mathbf{b})$ (when $a_{I(\mathbf{a},\mathbf{b})} \neq b_{I(\mathbf{a},\mathbf{b})}$, so $I \notin \mathcal{R}$), each term is strictly decreasing in $\pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a})$ because ϕ is. Thus, Fermi imitation dynamics satisfies condition (2). □

3.3 The cooperation ceiling

We now consider purely extrinsic population dynamics in games with two actions, C and D. Figure 3 sketches the argument that follows: in a social dilemma a defector pointwise dominates a cooperator, a relabelling of the two actions fixes a symmetric point at neutral payoffs, and together these bound the cooperation probability at one half. We first define a further classification for population dynamics:

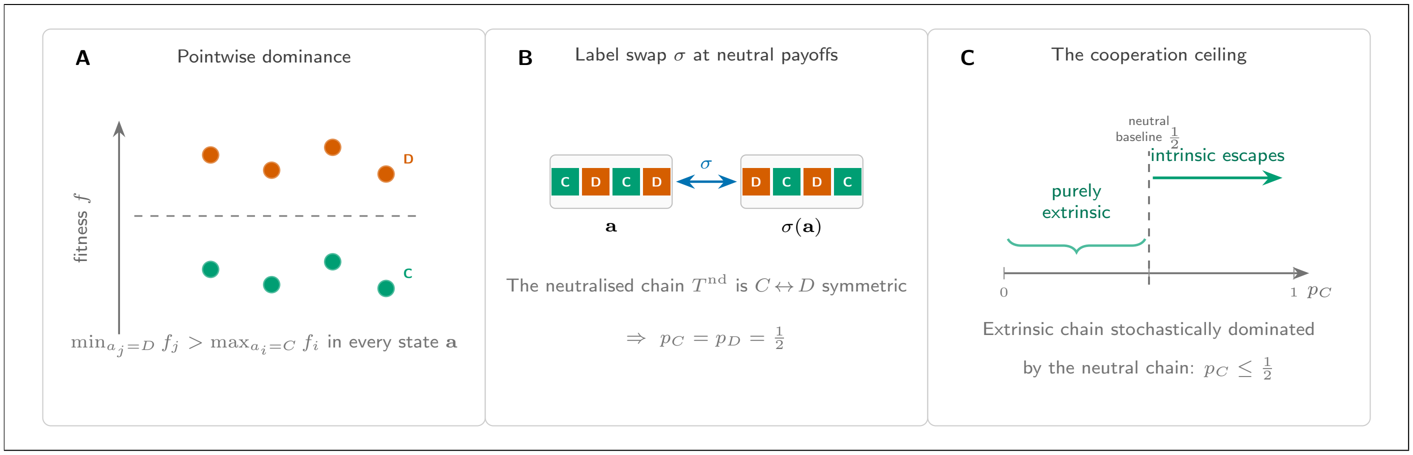

Figure 3: **Why purely extrinsic dynamics cannot exceed $p_C = \frac{1}{2}$. A,** In a social dilemma a defector pointwise dominates a cooperator: in any state, switching to D does not lower the payoff. **B,** At neutral payoffs the dynamic is unchanged by the label swap σ that exchanges C and D, fixing the symmetric point. **C,** Stochastic monotonicity then bounds the stationary cooperation probability at the neutral-drift baseline $p_C = \frac{1}{2}$.

Definition 2 (Neutrally monotone population dynamic)**.** *Let T be the transition matrix for a population dynamic on a game with two actions, $\{C, D\}$, and equip the state space $S = \{C, D\}^N$ with the componentwise partial ordering with $D < C$. Let T^{nd} denote T with all payoffs set to a common value (we denote by $v_{\mathbf{a}}$ the common value in state $\mathbf{a}$). We say that T defines a* neutrally monotone *population dynamic if:*

1. *$T^{nd}_{\mathbf{ab}}$ is independent of $v_{\mathbf{a}}$.*
2. *For every pair of states $\mathbf{a} \leq \mathbf{a}'$, there exist random variables $\mathbf{X}, \mathbf{X}'$ on a common probability space, with $\mathbf{X} \sim T^{\mathrm{nd}}(\mathbf{a}, \cdot)$ and $\mathbf{X}' \sim T^{\mathrm{nd}}(\mathbf{a}', \cdot)$, such that $\mathbf{X} \leq \mathbf{X}'$ almost surely.*

The two conditions ask that the neutralised chain T^{nd} is independent of the common payoff value (condition (1)) and is stochastically monotone (condition (2)). Neutral monotonicity thus captures configuration neutrality: under the same neutral payoffs, starting from a more cooperative profile cannot make the next state any less cooperative, and the chosen common value does not impact this neutrality. In a process which is not neutrally monotone, there could be a configuration-dependent pull toward a given action type at neutral payoffs, which is an inherent bias rather than truly neutral behaviour. It can be shown that this definition holds for both the Moran process, and Fermi imitation dynamics:

Theorem 3. *The Moran process and Fermi imitation dynamics are neutrally monotone under action-invariant mutation.*

Proof. We begin with the Moran process. Set $\pi_i(\mathbf{a}) = v$ for every player i gives $f_i(\mathbf{a}) = 1 - \varepsilon + \varepsilon v$ for every i, whence $T_{\mathbf{ab}} = \frac{|\mathcal{R}|(1-\varepsilon+\varepsilon v)}{N \cdot N(1-\varepsilon+\varepsilon v)}(1 - \sum_i \mu_{I(\mathbf{a},\mathbf{b}),i}) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),b_{I(\mathbf{a},\mathbf{b})}}}{N} = \frac{|\mathcal{R}|}{N^2}(1 - \sum_i \mu_{I(\mathbf{a},\mathbf{b}),i}) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),b_{I(\mathbf{a},\mathbf{b})}}}{N}$, which is independent of v. Therefore, condition (1) holds.

Under T^{nd}, as defined above, the Moran process samples a focal player i uniformly, a role model k uniformly from the population (since all fitnesses are equal), and a mutation outcome m from the action-invariant mutation distribution; player i then adopts a_k, or m if a mutation occurs. Couple the chains started at $\mathbf{a} \leq \mathbf{a}'$ on the common probability space generated by (i, k, m), applying the same draw to both. Coordinate i in the successor of $\mathbf{a}$ is a_k (or m); coordinate i in the successor of $\mathbf{a}'$ is a'_k (or the same m). Since $\mathbf{a} \leq \mathbf{a}'$ gives $a_k \leq a'_k$, and every other coordinate j is left unchanged with $a_j \leq a'_j$, the two successors $\mathbf{X}$ and $\mathbf{X}'$ satisfy $\mathbf{X} \leq \mathbf{X}'$ pointwise on this probability space, so condition (2) holds.

For Fermi imitation dynamics, if $\pi_i(\mathbf{a}) = v$ for every player i, then every payoff difference appearing in the Fermi formula is zero, so $\phi(\pi_I(\mathbf{a}) - \pi_j(\mathbf{a})) = \phi(0) = \frac{1}{2}$ in each summand, and $T_{\mathbf{ab}} = \frac{|\mathcal{R}|}{2N(N-1)}(1 - \sum_i \mu_{I(\mathbf{a},\mathbf{b}),i}) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),b_{I(\mathbf{a},\mathbf{b})}}}{N}$, independent of v. Therefore, condition (1) holds.

Under T^{nd}, the dynamic samples a focal player i uniformly, a role model k uniformly from the remaining $N-1$ players, an independent Bernoulli outcome with success probability $\phi(0) = \frac{1}{2}$, and a mutation outcome m from the action-invariant mutation distribution; player i adopts a_k on a success (or m if a mutation occurs) and otherwise keeps a_i. Couple the chains started at $\mathbf{a} \leq \mathbf{a}'$ on the common probability space generated by $(i, k, \mathrm{Bernoulli}, m)$, applying the same draw to both. On a mutation step, both successors set coordinate i to the same m. On a successful imitation step, they set coordinate i to a_k and a'_k respectively, with $a_k \leq a'_k$. On a no-update step, coordinate i retains $a_i \leq a'_i$. Every other coordinate is unchanged, so the two successors $\mathbf{X}$ and $\mathbf{X}'$ satisfy $\mathbf{X} \leq \mathbf{X}'$ pointwise on this probability space, and condition (2) holds. □

We will now show that there is a theoretic limit on the abundance of cooperators in any neutrally monotone purely extrinsic population dynamic. Before stating the theorem we record a comparison result for Markov chains on partially ordered spaces. The result is classical and we claim no novelty; we include a short proof for the reader's convenience, since its use in the proof of the theorem below may not be familiar to the reader.

Lemma 1. *Let P and Q be irreducible, aperiodic transition matrices on a finite partially ordered set $(S, \leq)$, with stationary distributions ρ^P and ρ^Q. Suppose that*

(i) $P(\mathbf{a}, \cdot) \preceq_{st} Q(\mathbf{a}, \cdot)$ for every $\mathbf{a} \in S$; and

(ii) Q is stochastically monotone: $\mathbf{a} \leq \mathbf{a}'$ implies $Q(\mathbf{a}, \cdot) \preceq_{st} Q(\mathbf{a}', \cdot)$.

Then $\rho^P \preceq_{st} \rho^Q$.

Proof. Kamae, Krengel, and O'Brien [25] prove that whenever two probability measures ν_1, ν_2 on a partially ordered Polish space satisfy $\nu_1 \preceq_{st} \nu_2$, there exist random variables $X_1 \sim \nu_1$ and $X_2 \sim \nu_2$ defined on a common probability space with $X_1 \leq X_2$ almost surely (a *monotone coupling*).

Using this, we construct a coupling of the two chains by induction on t. Start with any $X_0^P, X_0^Q \in S$ satisfying $X_0^P \leq X_0^Q$. Given $X_t^P \leq X_t^Q$, hypothesis (ii) gives $Q(X_t^P, \cdot) \preceq_{st} Q(X_t^Q, \cdot)$, which combined with (i) at $\mathbf{a} = X_t^P$ yields $P(X_t^P, \cdot) \preceq_{st} Q(X_t^Q, \cdot)$. Applying Kamae–Krengel–O'Brien to this pair of measures produces (X_{t+1}^P, X_{t+1}^Q) with the required marginals and $X_{t+1}^P \leq X_{t+1}^Q$ almost surely. By induction, $X_t^P \leq X_t^Q$ for every t. Since both chains are ergodic, their marginal laws converge to ρ^P and ρ^Q, and passing to the limit gives $\rho^P \preceq_{st} \rho^Q$. □

We now state the ceiling on the average abundance of cooperators in purely extrinsic population dynamics. Our result is comparable to that of [2], who studied the symmetric 2×2 game with payoff matrix

$$\begin{array}{c|cc} & A & B \\ \hline A & a & b \\ B & c & d \end{array} \tag{5}$$

and showed that, for a wide range of evolutionary processes with mutation, the condition $a(N-2) + bN > cN + d(N-2)$ makes strategy A more abundant than B at any mutation rate. This builds upon the work of [26,39], generalising results to arbitrary mutation and selection intensity. We now show that for neutrally monotone purely extrinsic population dynamics, pointwise domination of defection over cooperation is sufficient for defection to dominate cooperation in a heterogeneous population.

Theorem 4 (Extrinsic Ceiling)**.** *Let T define a neutrally monotone purely extrinsic process on an N-player game with 2 actions, C and D, such that in any given state $\mathbf{a}$, $\min_{j:a_j=D} \pi_j(\mathbf{a}) > \max_{i:a_i=C} \pi_i(\mathbf{a})$ (that is, any defector has greater payoff than any cooperator; equivalently, since fitness $f_i = 1 - \varepsilon + \varepsilon\pi_i$ is increasing in payoff, any defector has greater fitness than any cooperator), under action-invariant mutation.*

Then the mean proportion of cooperators in the steady state of T, denoted p_C, satisfies $p_C \leq \frac{1}{2}$.

Proof. Equip $S = \{C, D\}^N$ with the componentwise partial order, declaring $D < C$. For a transition matrix P, write $P(\mathbf{a}, \cdot) \preceq_{st} P(\mathbf{a}', \cdot)$ to mean that the one-step distribution from $\mathbf{a}$ under P is stochastically dominated by the one-step distribution from $\mathbf{a}'$ under P in this order. Let $T^{\text{nd}}_{\mathbf{ab}}$ denote the value of $T_{\mathbf{ab}}$ evaluated at $\pi_i(\mathbf{a}) = 0$ for every i; by neutral-monotonicity condition (1), $T^{\text{nd}}_{\mathbf{ab}}$ is equally the value of $T_{\mathbf{ab}}$ under any other choice of common payoff.

Step 1: $T(\mathbf{a}, \cdot) \preceq_{st} T^{\text{nd}}(\mathbf{a}, \cdot)$ for every $\mathbf{a} \in S$.

Set $v(\mathbf{a}) := \min_{a_j = D} \pi_j(\mathbf{a})$. In every mixed state $\mathbf{a}$, the payoff-dominance hypothesis implies $\pi_i(\mathbf{a}) < v(\mathbf{a})$ for every cooperator i and $\pi_j(\mathbf{a}) \geq v(\mathbf{a})$ for every defector j. Consider a C-gaining transition $\mathbf{a} \to \mathbf{b}$, so $b_{I(\mathbf{a},\mathbf{b})} = C$ and $\mathcal{R}$ is the set of current cooperators. Replacing every $\pi_i(\mathbf{a})$ with $v(\mathbf{a})$ raises all role-model payoffs and lowers all non-role-model payoffs, so by purely-extrinsic condition (2),

$$T_{\mathbf{ab}} \;\leq\; T_{\mathbf{ab}}\big|_{\pi_i(\mathbf{a}) = v(\mathbf{a}) \text{ for all } i}$$

with strict inequality whenever $\mathcal{R} \neq \emptyset$. By neutral-monotonicity condition (1), the right-hand side equals $T^{\text{nd}}_{\mathbf{ab}}$. The analogous argument with reversed inequality applies to D-gaining transitions. Hence $T(\mathbf{a}, \cdot) \preceq_{st} T^{\text{nd}}(\mathbf{a}, \cdot)$.

Step 2: T^{nd} is invariant under the $C \leftrightarrow D$ label swap.

Let $\sigma : S \to S$ be the map that swaps every C and D. By purely-extrinsic condition (3) and neutral-monotonicity condition (1), with common payoff 0, the only features of $(\mathbf{a}, \mathbf{b})$ that enter $T^{\text{nd}}_{\mathbf{ab}}$ are the mutation matrix and the set $\mathcal{R}$. Under action-invariant mutation the mutation matrix is σ-invariant; and $|\mathcal{R}|$ is σ-invariant because the set of players currently playing $b_{I(\mathbf{a},\mathbf{b})}$ in $\mathbf{a}$ has the same size as the set of players currently playing $\sigma(b_{I(\mathbf{a},\mathbf{b})})$ in $\sigma(\mathbf{a})$. Hence $T^{\text{nd}}_{\mathbf{ab}} = T^{\text{nd}}_{\sigma(\mathbf{a})\sigma(\mathbf{b})}$ for every transition, so the stationary distribution of T^{nd} is σ-invariant, giving $p_C^{(T^{\text{nd}})} = p_D^{(T^{\text{nd}})} = \frac{1}{2}$ [37].

Step 3: T^{nd} is stochastically monotone.

Neutral-monotonicity condition (2) provides, for every $\mathbf{a} \leq \mathbf{a}'$, a coupling $(\mathbf{X}, \mathbf{X}')$ on a common probability space with $\mathbf{X} \sim T^{\text{nd}}(\mathbf{a}, \cdot)$, $\mathbf{X}' \sim T^{\text{nd}}(\mathbf{a}', \cdot)$, and $\mathbf{X} \leq \mathbf{X}'$ almost surely. For any upper set $U \subseteq S$, $\{\mathbf{X} \in U\} \subseteq \{\mathbf{X}' \in U\}$, so $T^{\text{nd}}(\mathbf{a}, U) \leq T^{\text{nd}}(\mathbf{a}', U)$, giving $T^{\text{nd}}(\mathbf{a}, \cdot) \preceq_{st} T^{\text{nd}}(\mathbf{a}', \cdot)$.

Applying Lemma 1 to $P = T$ and $Q = T^{\text{nd}}$ (which satisfy (i) by Step 1 and (ii) by Step 3) yields $\rho^T \preceq_{st} \rho^{T^{\text{nd}}}$. Since n_C is monotone in the componentwise order,

$$p_C^{(T)} \;=\; \frac{1}{N}\,\mathbb{E}_T[n_C] \;\leq\; \frac{1}{N}\,\mathbb{E}_{T^{\text{nd}}}[n_C] \;=\; p_C^{(T^{\text{nd}})} \;=\; \frac{1}{2}.$$

□

Crucially, Theorem 4 shows that even if a player may increase their payoff by a switch $D \to C$ in any given state, pointwise dominance of D will still cause the process to favour D. This shows that a player's chosen strategy does not necessarily favour the action type which maximises their payoff in *either* the one-step or the long-run strategy distribution, instead favouring the strategy which is pointwise dominant over its peers. Even if there is no temptation to free-ride, players will still choose to defect based on the pointwise dominance of defectors in the state.

An example of this is the public goods game with $r > N$. This satisfies both pointwise dominance and a fitness increase for any player under the $D \to C$ swap. In such a game, there is no temptation to free ride; players will obtain a greater payoff by cooperating regardless of the population structure. The case of such a public goods game is considered in Figure 3 of [18], where a subset of a homogeneous population play a public goods game at each time step. The following section considers the heterogeneous public goods game as an example of Theorem 4.

3.3.1 Purely extrinsic population dynamics in the public goods game

To illustrate the results of section 3, we carry out a large parameter sweep across parameters of a public goods game [3, 21, 22, 23, 47]. In such a game, players choose to contribute some amount α_i. The total contribution is multiplied by some factor $r > 0$ and split between all players. We consider heterogeneous contributions and a homogeneous return.

The parameter sweep is done to the highest standards of reproducibility [43, 49, 52, 57, 58] and all data has been archived at [15] and is a further contribution of the work. In order to perform this sweep, the authors have developed an installable software library `ludics` (written in python). The archive of `ludics` is at [14] and the documentation can be found at [13]. The full development of the software can be found at [12]. The parameters considered, alongside their meanings in reference to the public goods game, are shown in Table 1.

Table 1: The set of measured parameters and boundary values. Some parameters are only applicable to certain processes. α_i is entirely controlled by the value M, which controls the maximum total contribution. The distribution of wealth remains the same for a given N, however as M increases, so too does the size of each player's contribution α_i. The value A is the *aspiration* considered in aspiration dynamics, in Section 4. We consider a homogeneous return r. The aspiration A depends on M, and the selection intensity ε on both M and r in addition to N than on N, and as such there are a large number of values per N for these parameters.

Parameter	Meaning	Minimum	Maximum	Number of Values per N
N	Number of players	2	8	N/A
r	Return on investment	$\frac{1}{2}$	$\frac{3N}{2}$	10
α_i	Contribution of player i	N/A	N/A	N/A
M	Maximum total contribution	N	$4N$	10
μ	Mutation value	0.001	0.1	4
ε	Selection intensity	0	$\varepsilon_{\max}$, see (6)	1000
β	Choice intensity	0	2	10
A	Aspiration	1	M	100

To keep every player's fitness positive, so that the Moran transition probabilities are well defined, the selection intensity is restricted to the interval $[0, \varepsilon_{\max}]$, where the upper bound depends on the return r and the largest contribution $\alpha_{\max} = \max_i \alpha_i$:

$$\varepsilon_{\max} = \begin{cases} \dfrac{0.99}{1 - \alpha_{\max}\left(\frac{r}{N} - 1\right)}, & r < N, \\ \dfrac{\alpha_{\max}\left(\frac{r}{N} - 1\right) + 1}{\alpha_{\max}\left(\frac{r}{N} - 1\right) + 2}, & r \geq N. \end{cases} \tag{6}$$

For each (M, r) we take ε on an evenly spaced grid of ten values in this interval.

Our results for the purely extrinsic dynamics are shown in Figure 4, and we explore Theorem 4 numerically in the context of the 8-player public goods game; the other values of N in our sweep are qualitatively identical. Panels (a) and (b) show this limit as a hard ceiling on p_C which no parameter set in our data is able to exceed. This includes parameter combinations which have high values of $r > N$, and so we see that despite full cooperation being the Nash equilibrium of the one shot game, and increasing the payoff of players in the one-step distribution, the pointwise dominance of defection's fitness over that of cooperation leads to defection becoming the dominant strategy in a repeated game according to a purely extrinsic population dynamic. This not only leads to a worse average payoff for all players in the population in the steady state, but also means that when $r > N$, worsening moves occur with a higher probability than changes of action types which improve a player's payoff. In other words, even when a switch $D \to C$ would always improve a player's payoff, the player will still prefer to remain a defector under a purely extrinsic population dynamic, due to the greater fitness of a defector in a given state.

In Fermi imitation dynamics, the invariance under r can be shown directly. In the public goods game the multiplied contribution $\frac{r}{N}\sum_{a_k=C}\alpha_k$ is shared equally by every player, so for any two players i and j in a state $\mathbf{a}$ this common term cancels in their payoff difference:

$$\pi_i(\mathbf{a}) - \pi_j(\mathbf{a}) = \alpha_j \mathbf{1}_{a_j=C} - \alpha_i \mathbf{1}_{a_i=C}, \tag{7}$$

which does not depend on r. As Fermi imitation dynamics depends on payoffs only through such pairwise differences, its transition matrix and steady state, and hence p_C, are invariant to r; this is confirmed in Figure 4(d). An increase in the choice intensity β has a negative effect upon p_C, as the value $\Delta(\pi)$ will always be negative for a $C \to D$ transition, and positive for a $D \to C$ transition. As the process with $\beta = 0$ gives us neutral drift, we see our result in practice, as moving towards a less rational process increases p_C in all empirical cases, up to the neutral-drift baseline. The notion of "rationality" in a purely extrinsic process is not directly linked to increasing a player's payoff. p_C in the Moran process increases with the value of r, however once again we see that it has a hard limit at $p_C = \frac{1}{2}$, as a given defector is always more likely to be chosen for reproduction than a given cooperator.

One interesting point is the invariance of the Moran process under the size of each player's contribution. The proportion of the state's total fitness given by that of defectors is:

$$\frac{qr\sum_{j=1}^{N}\alpha_j}{Nr\sum_{j=1}^{N}\alpha_j - \sum_{\mathbf{a}_l=C}\alpha_l} = \frac{qr\sum_{j=1}^{N}\alpha_j}{(Nr)\sum_{j=1}^{N}\alpha_j - \sum_{j=1}^{N}\alpha_j} = \frac{qr\sum_{j=1}^{N}\alpha_j}{(Nr-1)\sum_{j=1}^{N}\alpha_j} = \frac{qr}{Nr-1} \tag{8}$$

The analogous result for choosing a defector is given by $1 - \frac{qr}{Nr-1}$, which is once again invariant under α_i. Therefore, as Fermi imitation dynamics is invariant under the return parameter r, the Moran process is invariant under the contribution parameter α_i. We can observe this effect in Figure 4(c).

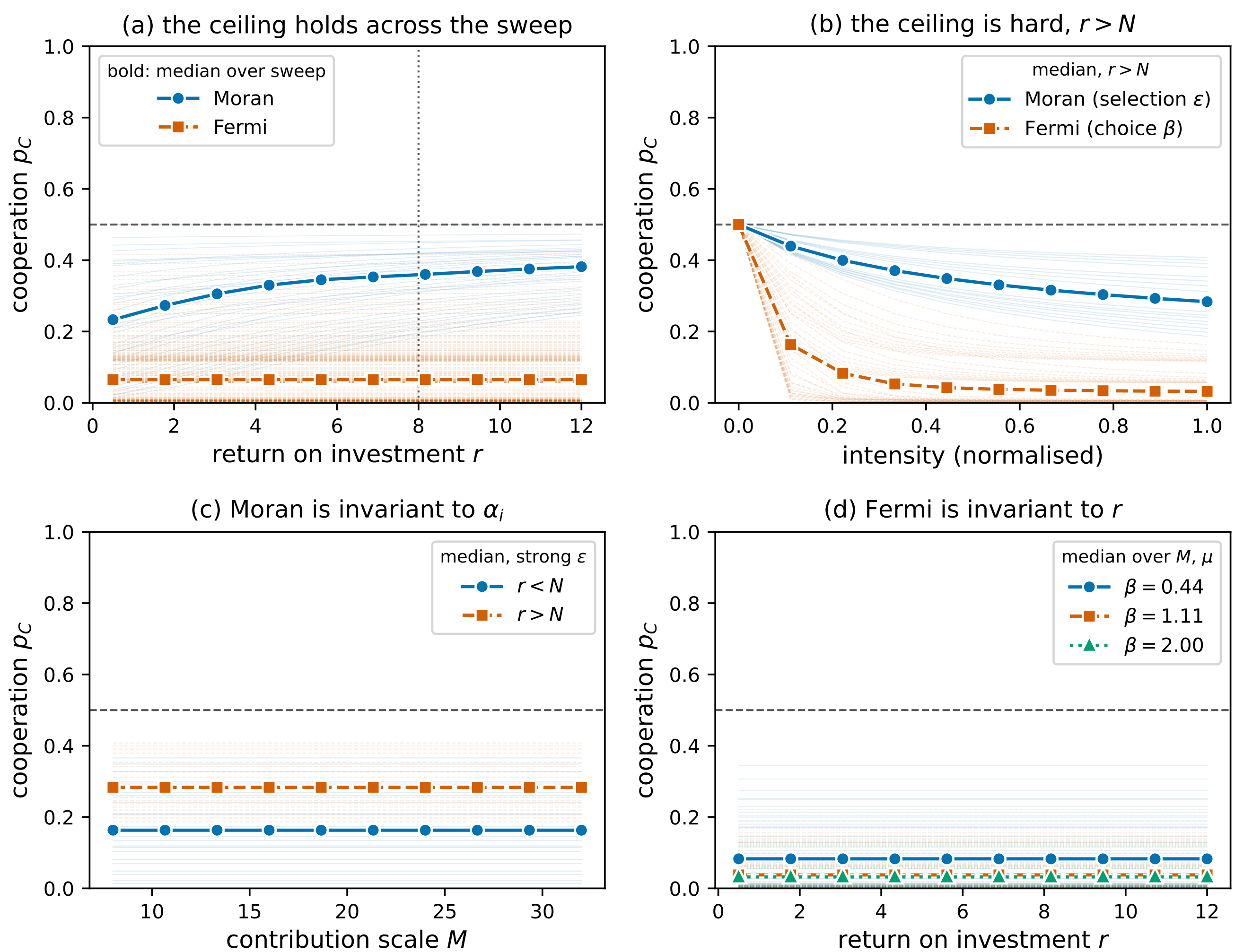

Figure 4: **The cooperation ceiling under purely extrinsic dynamics.** Throughout, $N = 8$, a linear contribution profile, and mutation $\mu = 0.05$; the dashed line marks the neutral-drift baseline $p_C = \frac{1}{2}$ and, in (a), the dotted line marks $r = N$. Faint curves show p_C trajectories across the sweep, coincident trajectories drawn once; in (a) they join parameter sets at equal selection- or choice-intensity rank across r, since the selection-intensity grid is rescaled with r (Equation (6)). Bold curves are the median over all parameter sets. **(a)** Cooperation p_C against the return r: the Moran process rises with r, with a dip just below $r = N$ where the selection-intensity bound is widest, but never exceeds $\frac{1}{2}$; Fermi imitation dynamics stays well below it. **(b)** With $r > N$, p_C against the normalised (dividing by the maximum value) selection intensity ε (Moran) and choice intensity β (Fermi); neither exceeds $\frac{1}{2}$, reaching it only as the intensity vanishes. **(c)** The Moran process is invariant to the contribution scale M, as derived in Equation (8). **(d)** Fermi imitation dynamics is invariant to the return r, as derived in Equation (7).

4 Intrinsic population dynamics

We have seen how purely extrinsic population dynamics only compare a player's payoffs to the payoff in the current state. We now see a classification of population dynamic where players instead compare their payoff to some attribute or payoff other than that of other players. We define a *purely intrinsic population dynamic* as follows:

Definition 3 (Purely intrinsic population dynamic)**.** *A population dynamic defined by a transition matrix T is said to be* purely intrinsic *if it satisfies the following conditions:*

1. *$T_{\mathbf{ab}}$ is a function of the payoffs of no player other than $I(\mathbf{a}, \mathbf{b})$ for $\mathbf{a} \neq \mathbf{b}$.*
2. *$T_{\mathbf{ab}}$ is decreasing in the payoff $\pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a})$ for $\mathbf{a} \neq \mathbf{b}$.*
3. *$T_{\mathbf{ab}}$ contains no term dependent on $a_{I(\mathbf{a},\mathbf{b})}$ or $b_{I(\mathbf{a},\mathbf{b})}$, except in the payoff function $\pi_{I(\mathbf{a},\mathbf{b})}$ and mutation parameters μ_{ij}.*

We now consider two purely intrinsic population dynamics: aspiration dynamics, which compares a player's payoff to their individual baseline aspiration, and introspection dynamics, which compares a player's payoff to their counterfactual payoff when they change action type.

4.1 Aspiration dynamics

Under aspiration dynamics, players choose to change action based on how far away their current payoff is from a target payoff, as shown in Figure 1C. It follows the algorithm:

1. A player i is chosen with probability $\frac{1}{N}$ to reconsider their strategy
2. This player changes strategy with a probability $\phi(\pi_i(\mathbf{a}) - A_i)$

Here, A_i is player i's *aspiration*, the payoff which they desire to make from a given game. A player will accept a new strategy with a higher than random probability if their current payoff is lower than their aspired payoff. This dynamic is traditionally paired with two action games [9, 10, 60], though it can be defined for games with 3 or more actions [59]. Aspiration dynamics has also often been combined with imitation dynamics to form a new population dynamic [4,30,42,54]. We consider the original form of aspiration dynamics defined in [10], with a homogeneous aspiration value A. The transition matrix is given by:

$$T_{\mathbf{ab}} = \begin{cases} \dfrac{1}{N}\,\phi(\pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a}) - A_{I(\mathbf{a},\mathbf{b})})\left(1 - \sum_{j=1}^{k} \mu_{I(\mathbf{a},\mathbf{b}),j}\right) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),\mathbf{b}_{I(\mathbf{a},\mathbf{b})}}}{N} & \text{if } \mathbf{b} \in \mathrm{Neb}(\mathbf{a}), \\ 0 & \text{if } \mathbf{b} \notin \mathrm{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b}. \end{cases} \tag{9}$$

This population dynamic spends a large amount of time in states where most players achieve their aspired payoffs. In this case, players will have a lower probability of switching strategy, compared to a state where players are underperforming in comparison to their target A. We now show that Aspiration dynamics is a purely intrinsic population dynamic:

Theorem 5 (Aspiration dynamics is a purely intrinsic population dynamic)**.** *Aspiration dynamics, as defined by Equation* (9)*, is a purely intrinsic population dynamic.*

Proof. It is clear from the definition of $T_{\mathbf{ab}}$ that aspiration dynamics satisfies conditions (1) and (3). It remains to show that aspiration dynamics is decreasing in the payoff $\pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a})$. Since $\phi(x)$ is decreasing, and $\pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a})$ appears only in the value $\pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a}) - A_i$. Therefore, setting $x = \pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a})$, we get that ϕ is decreasing in $\pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a})$, and therefore so is $T_{\mathbf{ab}}$. □

4.2 Introspection dynamics

Unlike the previous population dynamics we have seen, introspection dynamics [7, 8] allows players to consider the payoff they could have in other states. Under this population dynamic, players compare their current payoff with their counterfactual payoff upon changing action type. The process, as shown in Figure 1D, is given by the algorithm:

1. A player i is chosen with probability $\frac{1}{N}$ to reconsider their strategy
2. Player i randomly chooses another possible strategy, a_{il}, from the set of all possible strategies.
3. The chosen player replaces their strategy with probability $\phi(\Delta\pi_{I(\mathbf{a},\mathbf{b})})$, where $\Delta\pi_{I(\mathbf{a},\mathbf{b})} = \pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a}) - \pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{b})$ is the difference between the payoff of the player's current strategy and the payoff of the new strategy.

Introspection dynamics is well suited to a heterogeneous population, as players are able to consider whether a strategy will work for them, rather than for another player. As players check the payoff of a strategy in the next state, they never favour worsening moves in the one-step case, as the value $\phi(\Delta\pi_{I(\mathbf{a},\mathbf{b})})$ is greater than $\frac{1}{2}$ if and only if the change in a player's strategy increases their payoff. The transition matrix is given by:

$$T_{\mathbf{ab}} = \begin{cases} \dfrac{1}{N(k-1)}\,\phi(\Delta(\pi_{I(\mathbf{a},\mathbf{b})}))\left(1-\sum_{j=1}^{k}\mu_{I(\mathbf{a},\mathbf{b}),j}\right)+\frac{\mu_{I(\mathbf{a},\mathbf{b}),\mathbf{b}_{I(\mathbf{a},\mathbf{b})}}}{N} & \text{if } \mathbf{b}\in \mathrm{Neb}(\mathbf{a}),\\ 0 & \text{if } \mathbf{b}\notin \mathrm{Neb}(\mathbf{a}) \text{ and } \mathbf{a}\neq\mathbf{b},\\ 1-\sum_{\mathbf{b}\in S\setminus\{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a}=\mathbf{b}. \end{cases} \tag{10}$$

Introspection dynamics is aligned with classic static game theory, where a well informed and rational actor maximises their own payoff. We can contrast this with purely extrinsic population dynamics, which are the traditional method of modelling populations in evolutionary games, where actors are not typically well-informed and choose their next strategy according to fitnesses in the current population, without knowledge of the impact of their transition.

Theorem 6 (Introspection dynamics is a purely intrinsic population dynamic)**.** *Introspection dynamics, as defined by Equation* (10)*, is a purely intrinsic population dynamic.*

Proof. It is once again clear that introspection dynamics satisfies conditions (1) and (3). For condition (2), consider that $\phi(x)$ is decreasing. Setting $x = \pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a}) - \pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{b})$, we have our result. □

Unlike purely extrinsic population dynamics, introspection dynamics does not necessarily cancel out certain parameters of games, unless said parameter does not affect the difference in a player's payoff when playing different action types. For example, if we consider the public goods game, we can see that for introspection dynamics,

$$\Delta(\pi_{I(\mathbf{a},\mathbf{b})}) = \alpha_i\left(\frac{r}{N}-1\right), \tag{11}$$

which considers both α_i and r, and changes sign at $r = N$.

4.3 Intrinsic population dynamics in a public goods game

In Figure 5, we see that purely intrinsic population dynamics are able to exceed the ceiling on p_C which is imposed on purely extrinsic population dynamics. In the case of aspiration dynamics, p_C does not reach excessively low values as in the other processes we have discussed. This is because for all players to have a payoff $\pi_i > 0$, at least one player must be cooperating at all times. If no players cooperate, then no player will reach their target payoff threshold, and players will switch strategy with a high probability. The ceiling is not exceeded for $r < N$, as shown in Figure 5(a) and noted for a linear game in [10], but is exceeded at the threshold $r > N$. In this case, $\pi_i(\mathbf{a}) - A$ is greater when cooperating than defecting in a given population $\mathbf{a}_{-i}$ for all players, and thus the transition $D \to C$ will always occur with a higher probability than the reverse. For intermediate values of r, aspiration dynamics can achieve its lowest p_C, as with a low value of A, players are able to achieve their aspiration by free-riding in a state with fewer cooperators, and so there is no incentive to begin cooperating in such a state. Figure 5(d) shows how p_C depends on the aspiration level A. For $r > N$, p_C is non-monotone in A: it rises to a peak and then falls back towards $\frac{1}{2}$. Aspiration only discriminates between the two actions when A lies between the payoffs of a defector and a cooperator, or else both transitions $D \to C$ or $C \to D$ will be favoured or surpressed. A low A leaves defectors content and unwilling to switch, whereas an A above even the cooperator's payoff leaves every player dissatisfied, so switching becomes action-independent and the advantage of cooperation is washed out. The aspiration level is the biggest indicator of cooperation in a linear public goods game where a player's fitness is given by their expected return in a 2-player game. It has been shown previously that for a high aspiration level, cooperation emerges at a level near the neutral-drift baseline, but for a low aspiration level defectors dominate the population [10].

As the heterogeneous public goods game is an additive game [8], we can show that many of the analytical results of [16] about the structure of introspection dynamics in a public goods game hold empirically. Each player i cooperates with probability $p_i = \phi_i(\alpha_i(1-\frac{r}{N}))(1-2\mu)+\mu$, and p_C is given as a product measure [8]. Thus, we have that as the value $\Delta(\pi_{I(\mathbf{a},\mathbf{b})}) = \alpha_i(\frac{r}{N}-1)$ is positive for $r > N$ and negative for $r < N$, there is a sharp turning point at $r = N$ where introspection dynamics begins to favour cooperation, as shown in Figure 5(b). This gives an exact parameter value at which the ceiling of p_C is overcome.

Figure 5(c) shows the result that $\frac{\partial p_C}{\partial \alpha_i} > 0$ if and only if $r > N$, as the value $\alpha_i(\frac{r}{N}-1)$ becomes greater in this case, but becomes more negative if $r < N$. A greater contribution will increase a player's payoff if multiplied by a large r, but for a small r a larger contribution results in a greater loss to the player.

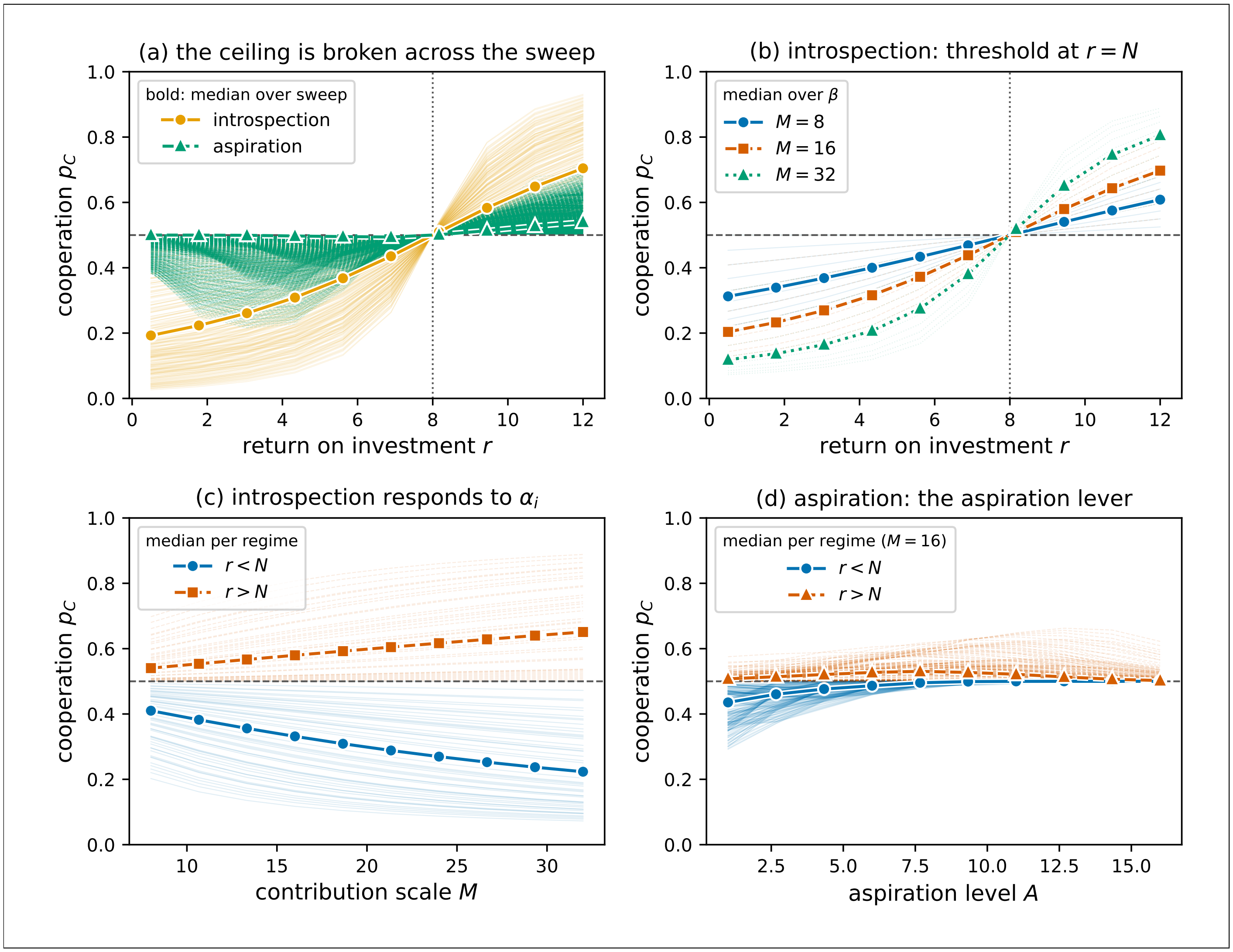

Figure 5: **Intrinsic dynamics exceed the cooperation ceiling.** Same population as Figure 4; faint curves show the p_C trajectory of every parameter set in the sweep (coincident trajectories are drawn once), and the bold curves their median taken over all parameter sets, with the dashed line at $p_C = \frac{1}{2}$ and the dotted line at $r = N$. **(a)** Cooperation p_C against the return r for introspection and aspiration dynamics; both exceed $\frac{1}{2}$ once $r > N$. **(b)** Introspection p_C against r at several contribution scales M, every curve crossing $\frac{1}{2}$ exactly at $r = N$: the threshold is set by the sign of the counterfactual payoff difference $\Delta(\pi_{I(\mathbf{a},\mathbf{b})})$ in Equation (11). **(c)** Introspection p_C against M: it increases with the contribution scale when $r > N$ and decreases when $r < N$. **(d)** Aspiration p_C against the aspiration level A at $M = 16$.

The same conditions also apply to the effect of β. This is because $\phi_i(\Delta(\pi_{I(\mathbf{a},\mathbf{b})}))$ is decreasing. Therefore, as β scales $\Delta(\pi_{I(\mathbf{a},\mathbf{b})})$, when $\Delta(\pi_{I(\mathbf{a},\mathbf{b})})$ is negative, $\frac{\partial p_C}{\partial \alpha_i} > 0$, and the inequality is reversed if the opposite is true. We acquire $\frac{\partial p_C}{\partial \alpha_i} < 0$ at exactly the value $r > N$, and thus we have our result. These results are proven in [16], but have been shown empirically for our large data sweep in Figure 5.

Pooling the entire parameter sweep makes the contrast between intrinsic and extrinsic dynamics explicit. Figure 6 collects every parameter combination with $\mu > 0$, across all N, contributions, returns, intensities and aspirations. No purely extrinsic parameter set anywhere in the sweep exceeds the neutral-drift baseline $p_C = \frac{1}{2}$, and the largest p_C attained is pinned at exactly $\frac{1}{2}$ for every N. The intrinsic dynamics, by contrast, place a substantial fraction of their mass above the ceiling, and this fraction switches on precisely at $r = N$ (Table 2).

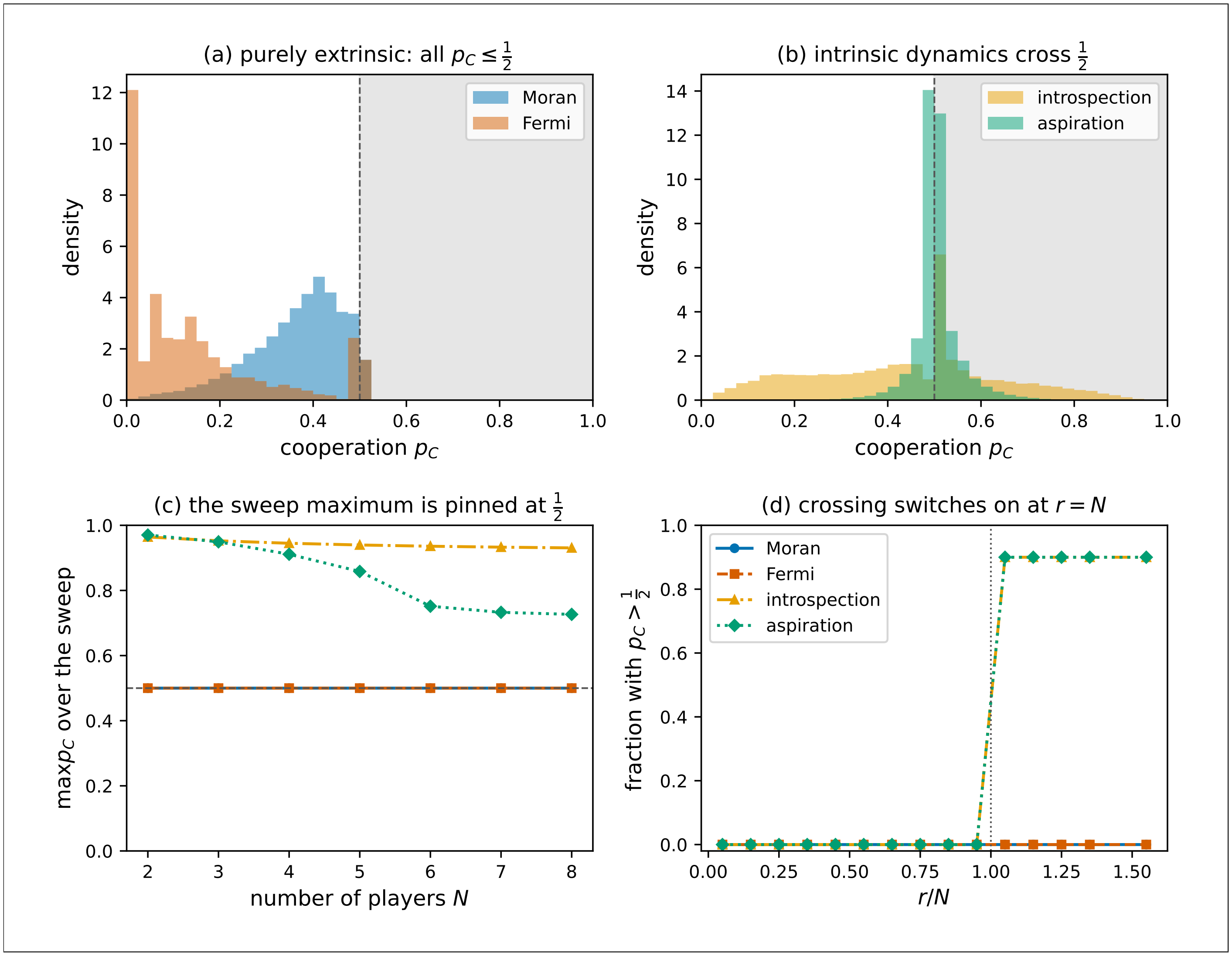

Figure 6: **The cooperation ceiling across the entire sweep.** Each panel pools every parameter combination with $\mu > 0$. **(a)** The distribution of p_C for the purely extrinsic dynamics lies entirely at or below the neutral-drift baseline $p_C = \frac{1}{2}$ (shaded region). **(b)** The intrinsic dynamics place mass above $\frac{1}{2}$. **(c)** The maximum p_C attained anywhere in the sweep, as a function of N: for the extrinsic dynamics it is pinned at exactly $\frac{1}{2}$ at every N. **(d)** The fraction of parameter sets with $p_C > \frac{1}{2}$ as a function of r/N: it is zero for the extrinsic dynamics at every return, and for the intrinsic dynamics switches on precisely at $r = N$.

Table 2: Percentage of parameter sets whose steady state crosses the cooperation ceiling, $p_C > \frac{1}{2}$, by dynamic and return regime, over the full sweep with $\mu > 0$.

dynamic	parameter sets	all	$r < N$	$r > N$
Moran	28,000	0.0%	0.0%	0.0%
Fermi	28,000	0.0%	0.0%	0.0%
introspection	28,000	36.0%	0.0%	90.0%
aspiration	280,000	36.0%	0.0%	90.0%

5 Robustness to population size

The exact steady-state analysis and the parameter sweep so far have used small populations, for which the transition matrix can be formed explicitly. Since the size of the state space grows as 2^N, this is infeasible for large N. To confirm that the cooperation ceiling and its breaking are not artefacts of small populations, we estimate p_C for populations as large as $N = 100$ by simulating the Markov chain directly with [14] and averaging the cooperator fraction over an ergodic run with mutation $\mu = 0.05$.

Figure 7 shows that the conclusions of the paper survive at $N = 100$. Panel 7A shows that the purely extrinsic dynamics remain at or below the neutral-drift baseline $p_C = \frac{1}{2}$ for all returns; introspection dynamics rises clearly through $\frac{1}{2}$ at $r = N$, while aspiration dynamics stays close to it. Panel 7B shows the cooperation probability at a high return ($r/N = 1.3$) for each population size: the extrinsic dynamics stay well below $\frac{1}{2}$ across the whole range $10 \leq N \leq 100$, introspection lies clearly above it, and aspiration sits just above $\frac{1}{2}$ with a margin that narrows as N grows. Panel 7C takes introspection at a different choice intensity β for each of $N \in \{10, 50, 100, 200\}$: the curves differ in steepness, yet every one of them crosses $\frac{1}{2}$ at exactly $r = N$, so the threshold is robust to both the population size and the choice intensity. Panel 7D validates the simulator against the exact steady state for $2 \leq N \leq 8$: the simulated estimates bracket the exact value for all four dynamics. The extrinsic ceiling is therefore unbroken at every population size, and an intrinsic dynamic continues to exceed it for large N.

6 Conclusion

We have introduced and analysed a framework for heterogeneous evolutionary games under four distinct population dynamics in two categories, as shown in Figure 1. We have defined the categorisation of a *purely extrinsic population dynamic*, and given rigorous formulations of two widely used examples: the Moran process and Fermi imitation dynamics. We have shown that dynamics under this classification have a hard ceiling on the probability of cooperation p_C in social dilemmas where a defector *always* outperforms a cooperator in a given state. In such a game, purely extrinsic population dynamics are unable to exceed the neutral-drift baseline. We then give formulations of two purely intrinsic population dynamics: aspiration dynamics in which players change strategy based on their performance in relation to a target payoff, and introspection dynamics in which players compare their current payoff with their counterfactual payoff. We have also shown empirical results for the public goods game with heterogeneous contributions which confirm the extrinsic ceiling, as well as the ability for purely intrinsic population dynamics to exceed it.

Further work may consider populations in which the choice between intrinsic and extrinsic update rules is itself a heterogeneous attribute of the players, in the spirit of the combination of Moran and Fermi processes in [32]. Figure 8(a) illustrates this for a population that mixes Moran and introspection players: once the return exceeds the threshold $r = N$ the population can exceed the neutral-drift baseline, though the number of intrinsic players required depends on the relative strength of the two steps. A stronger extrinsic selection intensity ε, or a weaker choice Moran and introspection players: once the return exceeds the threshold $r = N$ intensity β, demands more intrinsic players before p_C crosses the baseline. For $r < N$ the same intrinsic players instead suppress cooperation. The heterogeneity considered in this paper is also only in the players' contributions, whereas the introspection threshold acts at the level of each player's own return. Figure 8(b) shows that under a heterogeneous return r_i, cooperation crosses the ceiling once enough players sit above the $r = N$ threshold. The interplay between such heterogeneous returns, contributions and update rules is a natural direction for further study. Finally, we only consider the ceiling of p_C in two-action games for purely extrinsic population dynamics. It is entirely possible that there is a formulation of a k-action game such that a ceiling is posed on the abundance of a certain action. This may be a focus of future work, extending the limitation of purely extrinsic population dynamics to a larger set of games.

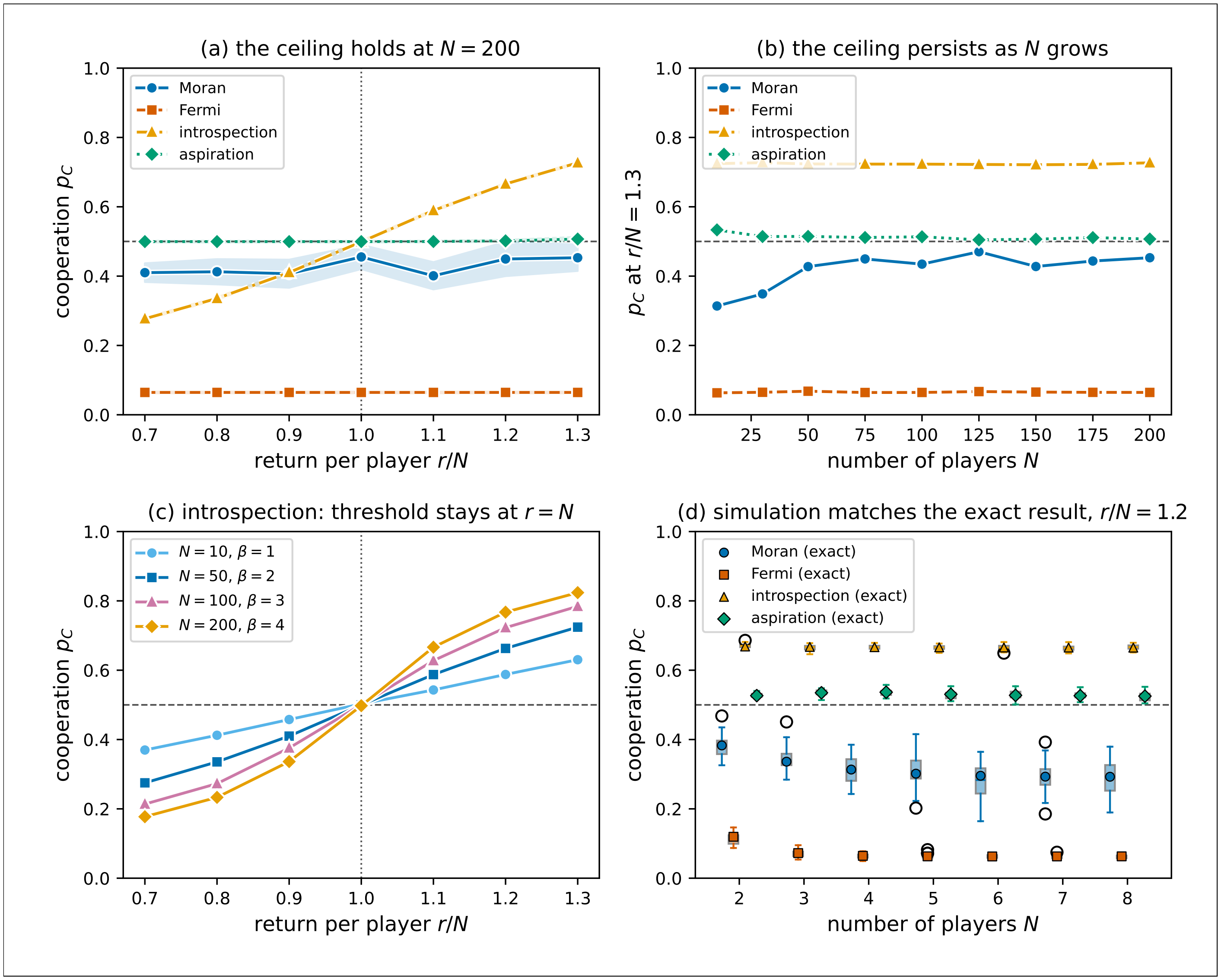

Figure 7: **The cooperation ceiling persists for large populations.** Cooperation p_C is estimated by simulating the Markov chain and averaging the cooperator fraction over the run. Throughout, mutation $\mu = 0.05$, a linear contribution profile of scale $M = 2N$, selection intensity $\varepsilon = 0.5$ for the Moran process, choice intensity $\beta = 2$ for Fermi, introspection and aspiration dynamics, and aspiration level $A = 0.7\,M$; bands and boxes show the variation over independent seeds. **A,** p_C against r/N at $N = 100$: the extrinsic dynamics (Moran, Fermi) stay at or below $\frac{1}{2}$; introspection crosses it at $r = N$, while aspiration only marginally exceeds it. **B,** p_C at a high return ($r/N = 1.3$), against the population size N: below $\frac{1}{2}$ for the extrinsic dynamics at every N, clearly above it for introspection, and just above it for aspiration. **C,** Introspection p_C against r/N, each N at a different choice intensity β: the curves differ in shape but all cross $\frac{1}{2}$ at $r = N$. **D,** Validation against the exact steady state for $2 \le N \le 8$ at $r/N = 1.2$: the boxes show the simulated estimates over seeds and the markers the exact values, for all four dynamics.

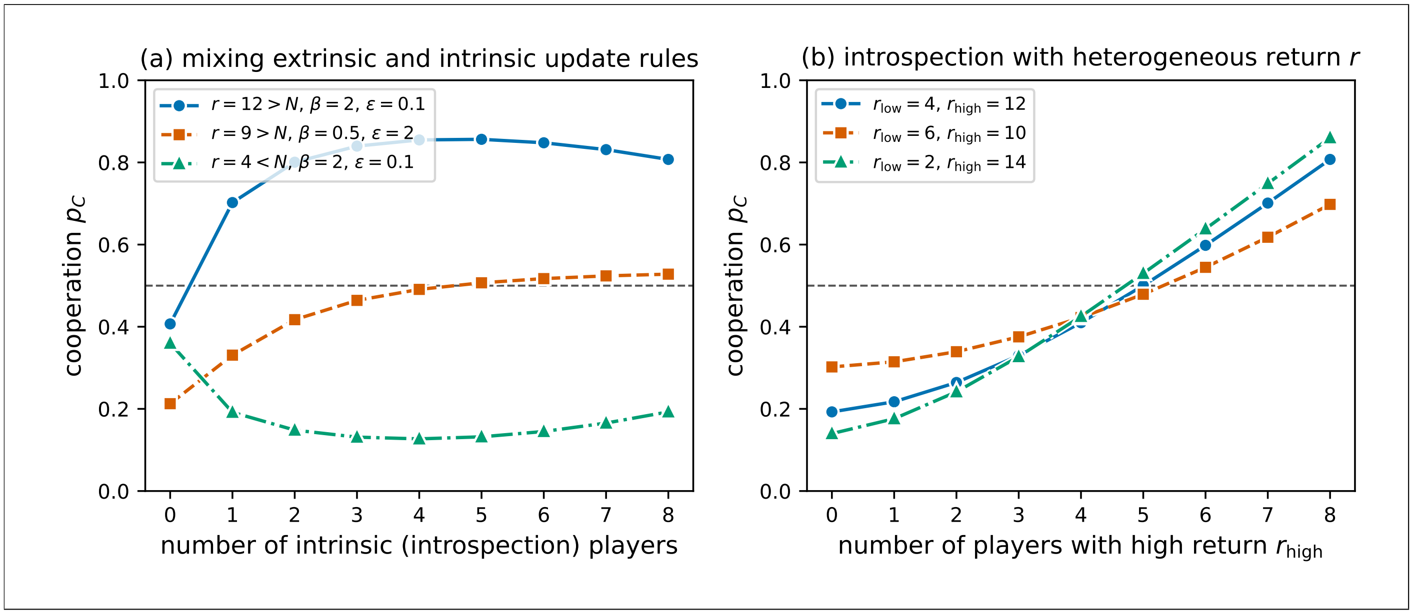

Figure 8: **Heterogeneous update rules and heterogeneous returns.** Both panels are computed exactly from the steady state of the corresponding transition matrix, with $N = 8$, a linear contribution profile of scale $M = 16$, and mutation $\mu = 0.05$; the dashed line marks the neutral-drift baseline $p_C = \frac{1}{2}$. **(a)** A population that mixes Moran (extrinsic) and introspection (intrinsic) players, against the number of introspection players: with $r > N$ the population can exceed the baseline, but the number of intrinsic players required grows as the extrinsic selection intensity ε increases or the choice intensity β falls; with $r < N$ the intrinsic players instead suppress cooperation. **(b)** Introspection dynamics with a heterogeneous return, against the number of players receiving the high return r_{high}; cooperation crosses the ceiling once enough players sit above the threshold $r = N$.

7 Acknowledgements

Harry Foster's research was supported by EPSRC grant EP/Z535126/1.

References

[1] Madjid Amir and Siegfried K. Berninghaus. Another approach to mutation and learning in games. *Games and Economic Behavior*, 14:19–43, 1996.

[2] Tibor Antal, Martin A. Nowak, and Arne Traulsen. Strategy abundance in 2×2 games for arbitrary mutation rates. *Journal of Theoretical Biology*, 257(2):340–344, 2009.

[3] Marco Archetti and István Scheuring. Review: Game theory of public goods in one-shot social dilemmas without assortment. *Journal of Theoretical Biology*, 299:9–20, 2012. Evolution of Cooperation.

[4] Md. Rajib Arefin and Jun Tanimoto. Evolution of cooperation in social dilemmas under the coexistence of aspiration and imitation mechanisms. *Phys. Rev. E*, 102:032120, Sep 2020.

[5] Robert Axelrod. Effective choice in the prisoner's dilemma. *The Journal of Conflict Resolution*, 24(1):3–25, 1980.

[6] Immanuel Bomze and Christina Pawlowitsch. One-third rules with equality: Second-order evolutionary stability conditions in finite populations. *Journal of theoretical biology*, 254:616–20, 07 2008.

[7] Marta Couto, S Giaimo, and Christian Hilbe. Introspection dynamics: A simple model of counterfactual learning in asymmetric games. *New Journal of Physics*, 24, 06 2022.

[8] Marta Couto and Saptarshi Pal. Introspection dynamics in asymmetric multiplayer games. *Dynamic Games and Applications*, 13:1–30, 09 2023.

[9] Jia Du, Bin Wu, and Long Wang. Aspiration dynamics in structured population acts as if in a well-mixed one. *Scientific Reports*, 5:8014, 2015.

[10] Jinming Du, Bin Wu, Philipp M. Altrock, and Long Wang. Aspiration dynamics of multi-player games in finite populations. *Journal of The Royal Society Interface*, 11(99):20140778, 2014.

[11] Lucas S. Flores, Mendeli H. Vainstein, Heitor C. M. Fernandes, and Marco A. Amaral. Heterogeneous contributions can jeopardize cooperation in the public goods game. *Phys. Rev. E*, 108:024111, Aug 2023.

[12] Harry Foster and Vince Knight. ludics. `https://github.com/hefos/ludics`, 2026.

[13] Harry Foster and Vince Knight. ludics documentation. `https://hefos.github.io/ludics/`, 2026.

[14] Harry Foster and Vincent Knight. hefos/ludics (v0.3.2), 2026.

[15] Harry Foster, Vincent Knight, and Sebastian Krapohl. Data for: The ceiling of cooperation in extrinsic update rules, 2026. TODO: archive the data on Zenodo and insert the DOI.

[16] Harry Foster, Vincent A. Knight, and Sebastian Krapohl. Introspection dynamics with mutation in additive games. 2026.

[17] Drew Fudenberg, Martin A. Nowak, Christine Taylor, and Lorens A. Imhof. Evolutionary game dynamics in finite populations with strong selection and weak mutation. *Theoretical Population Biology*, 70(3):352–363, 2006.

[18] Alexander G. Ginsberg and Feng Fu. Evolution of cooperation in public goods games with stochastic opting-out. *Games*, 10(1), 2019.

[19] Frank Harary, John P. Hayes, and Horng-Jyh Wu. A survey of the theory of hypercube graphs. *Computers & Mathematics with Applications*, 15(4):277–289, 1988.

[20] Garrett Hardin. The tragedy of the commons. *Science*, 162(3859):1243–1248, 1968.

[21] Christoph Hauert, Silvia De Monte, Josef Hofbauer, and Karl Sigmund. Volunteering as Red Queen mechanism for cooperation in public goods games. *Science*, 296(5570):1129–1132, 2002.

[22] Christoph Hauert, Nina Haiden, and Karl Sigmund. The dynamics of public goods. *Discrete and Continuous Dynamical Systems - B*, 4(3):575–587, 2004.

[23] Oliver P. Hauser, Christian Hilbe, Krishnendu Chatterjee, and Martin A. Nowak. Social dilemmas among unequals. *Nature*, 572(7770):524–527, 2019.

[24] Josef Hofbauer and Karl Sigmund. *Evolutionary Games and Population Dynamics*. Cambridge University Press, 1998.

[25] Teturo Kamae, Ulrich Krengel, and George L. O'Brien. Stochastic inequalities on partially ordered spaces. *The Annals of Probability*, 5(6):899–912, 1977.

[26] Michihiro Kandori, George J. Mailath, and Rafael Rob. Learning, mutation, and long run equilibria in games. *Econometrica*, 61(1):29–56, 1993.

[27] Vincent Knight, Owen Campbell, Marc Harper, Karol M. Langner, Jonathan Campbell, Thomas Campbell, Alex Carney, Martin Chorley, Cameron Davidson-Pilon, Kristian Glass, Nikoleta Glynatsi, Tomáš Ehrlich, Martin Jones, Georgios Koutsovoulos, Jamie Muller, Geraint Ng, Remi Noble, Nikolay Ryzhikov, Jessica Standen, Jon Taylor, Michail Theodorakis, Yingrong Wu, and Jan Žufan. Evolution reinforces cooperation with the emergence of self-recognition mechanisms: An empirical study of strategies in the Moran process for the iterated prisoner's dilemma. *PLOS ONE*, 13(10):e0204981, 2018.

[28] Vincent Knight, Marc Harper, Nikoleta E. Glynatsi, and Jonathan Gillard. Recognising and evaluating the effectiveness of extortion in the iterated prisoner's dilemma. *PLOS ONE*, 19(7):1–17, 07 2024.

[29] Chuang Lei, Te Wu, Jian-Yuan Jia, Rui Cong, and Long Wang. Heterogeneity of allocation promotes cooperation in public goods games. *Physica A: Statistical Mechanics and its Applications*, 389(21):4708–4714, 2010.

[30] Xuesong Liu, Mingfeng He, Yibin Kang, and Qiuhui Pan. Aspiration promotes cooperation in the prisoner's dilemma game with the imitation rule. *Phys. Rev. E*, 94:012124, Jul 2016.

[31] Xuesong Liu, Mingfeng He, Yibin Kang, and Qiuhui Pan. Fixation of strategies with the moran and fermi processes in evolutionary games. *Physica A: Statistical Mechanics and its Applications*, 484:336–344, 2017.

[32] Xuesong Liu, Qiuhui Pan, Yibin Kang, and Mingfeng He. Fixation probabilities in evolutionary games with the moran and fermi processes. *Journal of Theoretical Biology*, 364:242–248, 2015.

[33] Xiaojian Ma, Ji Quan, and Xianjia Wang. Effect of reputation-based heterogeneous investment on cooperation in spatial public goods game. *Chaos, Solitons & Fractals*, 152:111353, 2021.

[34] Manfred Milinski, Torsten Röhl, and Jochem Marotzke. Cooperative interaction of rich and poor can be catalyzed by intermediate climate targets. *Climatic Change*, 109(3-4):807–814, 2011.

[35] Manfred Milinski, Ralf D. Sommerfeld, Hans-Jürgen Krambeck, Floyd A. Reed, and Jochem Marotzke. The collective-risk social dilemma and the prevention of simulated dangerous climate change. *Proceedings of the National Academy of Sciences*, 105(7):2291–2294, 2008.

[36] P. A. P. Moran. Random processes in genetics. *Mathematical Proceedings of the Cambridge Philosophical Society*, 54(1):60–71, 1958.

[37] MARTIN A. NOWAK. *Evolutionary Dynamics: Exploring the Equations of Life*. Harvard University Press, 2006.

[38] Martin A. Nowak and Robert M. May. Evolutionary games and spatial chaos. *Nature*, 359:826–829, 1992.

[39] Martin A. Nowak, Akira Sasaki, Christine Taylor, and Drew Fudenberg. Emergence of cooperation and evolutionary stability in finite populations. *Nature*, 428:646–650, 2004.

[40] Hisashi Ohtsuki, Pedro Bordalo, and Martin A. Nowak. The one-third law of evolutionary dynamics. *Journal of Theoretical Biology*, 249(2):289–295, 2007.

[41] Matjaž Perc, Jesús Gómez-Gardeñes, Attila Szolnoki, Luis M. Floría, and Yamir Moreno. Evolutionary dynamics of group interactions on structured populations: a review. *Journal of the Royal Society Interface*, 10(80):20120997, 2013.

[42] Ji Quan, Yawen Zhou, Xianjia Wang, and Jian-Bo Yang. Evidential reasoning based on imitation and aspiration information in strategy learning promotes cooperation in optional spatial public goods game. *Chaos, Solitons & Fractals*, 133:109634, 2020.

[43] Geir Kjetil Sandve et al. Ten simple rules for reproducible computational research. *PLoS Computational Biology*, 9(10):e1003285, 2013.

[44] Francisco C. Santos and Jorge M. Pacheco. Scale-free networks provide a unifying framework for the emergence of cooperation. *Physical Review Letters*, 95:098104, 2005.

[45] Francisco C. Santos and Jorge M. Pacheco. Risk of collective failure provides an escape from the tragedy of the commons. *Proceedings of the National Academy of Sciences*, 108(26):10421–10425, 2011.

[46] Francisco C. Santos, Marta D. Santos, and Jorge M. Pacheco. Social diversity promotes the emergence of cooperation in public goods games. *Nature*, 454:213–216, 2008.

[47] Karl Sigmund. *The Calculus of Selfishness:*. Princeton University Press, 2010.

[48] William J. Stewart. *Probability, Markov Chains, Queues, and Simulation: The Mathematical Basis of Performance Modeling*. Princeton University Press, 2009.

[49] Victoria Stodden, Jennifer Seiler, and Zhaokun Ma. Enhancing reproducibility for computational methods. *PNAS*, 115(11):2561–2570, 2018.

[50] György Szabó and Gábor Fáth. Evolutionary games on graphs. *Physics Reports*, 446:97–216, 2007.

[51] György Szabó and Csaba Tőke. Evolutionary prisoner's dilemma game on a square lattice. *Phys. Rev. E*, 58:69–73, Jul 1998.

[52] The Turing Way Community. *The Turing Way: A Handbook for Reproducible, Ethical and Collaborative Data Science.* Zenodo, 2022. Zenodo. DOI: 10.5281/zenodo.3233853.

[53] Arne Traulsen, Martin A. Nowak, and Jorge M. Pacheco. Stochastic dynamics of invasion and fixation. *Physical Review E*, 74(1):011909, 2006.

[54] Si-Yi Wang, Qing-Lian Wang, Xiao-Wei Zhang, and Rui-Wu Wang. Evolutionary cooperation dynamics of combining imitation and super-rational aspiration induced strategy updating. *Applied Mathematics and Computation*, 456:128134, 2023.

[55] Zhenyu Wang and Nigel Goldenfeld. Theory of cooperation in a micro-organismal snowdrift game. *Phys. Rev. E*, 84:020902, Aug 2011.

[56] Joshua S. Weitz, Ceyhun Eksin, Keith Paarporn, Sam P. Brown, and William C. Ratcliff. An oscillating tragedy of the commons in replicator dynamics with game-environment feedback. *bioRxiv*, 2016.

[57] Greg Wilson, D. A. Aruliah, C. Titus Brown, Neil P. Chue Hong, Matt Davis, Richard T. Guy, Steven H. D. Haddock, Kathryn D. Huff, Ian M. Mitchell, Mark D. Plumbley, Ben Waugh, Ethan P. White, and Paul Wilson. Best practices for scientific computing. *PLOS Biology*, 12(1):e1001745, jan 2014.

[58] Greg Wilson et al. Good enough practices in scientific computing. *PLoS Computational Biology*, 13(6):e1005510, 2017.

[59] Jun Zhang, Bin Hu, Yi Jie Huang, Zheng Hong Deng, and Tao Wu. The evolution of cooperation affected by aspiration-driven updating rule in multi-games with voluntary participation. *Chaos, Solitons & Fractals*, 139:110067, 2020.

[60] Lei Zhou, Bin Wu, Jia Du, et al. Aspiration dynamics generate robust predictions in heterogeneous populations. *Nature Communications*, 12:3250, 2021.